School of Science, Computing and Emerging Technologies

Applied Machine Learning

Lab 6. Object Detection with YOLO

Overview

In this lab, you will learn how to use YOLO (You Only Look Once) for object detection. The implementation used is provided by Ultralytics (https://docs.ultralytics.com/). You will evaluate a pretrained model and then fine-tune it on your own dataset to compare performance.

Dataset

You are required to select a small-scale object detection dataset (approximately 100–300 images) with a train/val/test split and 2–5 classes. The dataset can be sourced from the following platforms:

- Roboflow (https://roboflow.com/), or
- Kaggle (https://www.kaggle.com/)

The dataset must be in YOLO format.

Download the dataset and examine its folder structure and annotation format. Please include the link to your dataset in your submitted source code.

Use a Pretrained Model

Refer to the following YOLO tutorial:

https://colab.research.google.com/github/ultralytics/ultralytics/blob/main/examples/tutorial.ipynb

Use a pretrained model (e.g., yolo26n.pt) and evaluate it on your dataset's test set. Report the following metrics: mAP50, mAP50–95, precision, and recall.

Fine-tune the Model

Fine-tune the pretrained model using your dataset. This process adapts the model to your specific dataset. Example code:

```
model.train(
   data="dataset/data.yaml",
   epochs=30,
   imgsz=640,
   freeze=10)
```

After training, evaluate the model again on the same test set and compare the results with those of the pretrained model.

Questions

Answer the following questions in a Word document:

1. Describe the structure of a YOLO dataset.
2. Explain the YOLO label format and why coordinates are normalised.
3. What is a pretrained model, and why can it perform reasonably well before training?
4. What do IoU, Precision, Recall, and mAP measure in object detection?
5. What is the difference between mAP50 and mAP50–95?
6. Compare the pretrained and fine-tuned model results, and explain why performance improved or decreased.

Submission

Submit your source code and answers to the questions via Canvas. You are also required to demonstrate your work to the tutor during the lab session.